

Supervised Learning

Supervised learning is the most common type of machine learning and is responsible for most of its economic value today. It's all about learning a mapping from an **input (x)** to an **output (y)** based on labeled examples.

How it Works 🧠

The core idea is to "supervise" the algorithm by giving it a dataset that includes the "right answers."

1. **Training:** The algorithm is fed a large amount of training data, where each data point consists of an input x and its corresponding correct output label y .
2. **Learning:** By seeing many (x, y) pairs, the algorithm learns the underlying relationship between the inputs and outputs.
3. **Prediction:** Once trained, the model can be given a **new input x** that it has never seen before and produce a prediction, or a good guess, for the output y .

Examples of Supervised Learning

Application	Input (x)	Output (y)
Spam filtering	An email	Spam (1) or Not Spam (0)
Speech recognition	An audio clip	Text transcript
Machine translation	A sentence in English	The sentence in Spanish
Online advertising	Ad info + user info	Will the user click? (Yes/No)
Self-driving cars	Image from a camera	Position of other cars
Visual inspection	Picture of a phone	Defect (1) or No Defect (0)

Types of Supervised Learning: Regression

One of the two major types of supervised learning is called **regression**.

- **Goal:** To predict a **continuous numerical value**. This means the output can be any number within a range.
- **Example: Housing Price Prediction**

- You provide the model with a dataset of houses, where the input x is the **size of the house** (e.g., in square feet) and the output y is its **selling price**.
- The algorithm learns the relationship by fitting a line or a curve to this data.
- It can then predict a price for a house size it has never seen before.